Implicit Differentiation and Inverse Functions

Section 1: First Derivative by Implicit Differentiation

1 Find $\frac{dy}{dx}$ for the circle $x^2 + y^2 = 25$.

a. $-\frac{x}{y}$

b. $\frac{x}{y}$

c. $-\frac{y}{x}$

d. $\frac{y}{x}$

2 Find $\frac{dy}{dx}$ for the curve $x^3 + y^3 = 6xy$.

a. $\frac{2y - x^2}{y^2 - 2x}$

b. $\frac{x^2 - 2y}{2x - y^2}$

c. $\frac{2x - y^2}{x^2 - 2y}$

d. $\frac{x^2 + y^2}{6}$

3 Find the slope of the tangent line to $x^2y + y^2 = 5$ at the point $(2, 1)$.

a. $-\frac{2}{3}$

b. $-\frac{4}{5}$

c. $-\frac{4}{6}$

d. -1

Section 2: Second Derivatives Implicitly

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4 For $x^2 + y^2 = 16$, what is $\frac{d^2y}{dx^2}$ expressed solely in terms of y?

a. $-\frac{16}{y^3}$

b. $\frac{16}{y^3}$

c. $-\frac{1}{y^2}$

d. $-\frac{4}{y^3}$

5 At the point $(0, 2)$ on the ellipse $4x^2 + y^2 = 4$, what is $\frac{d^2y}{dx^2}$?

a. -2

b. 2

c. -4

d. 0

6 What is the sign of the concavity at $(3, 4)$ on the circle $x^2 + y^2 = 25$?

a. Negative (concave down)

b. Positive (concave up)

c. Zero (inflection point)

d. Undefined

Section 3: Derivatives of Inverse Functions

7 If g is the inverse of f, which formula gives $g'(a)$ where $f(b) = a$?

a. $g'(a) = \frac{1}{f'(b)}$

b. $g'(a) = f'(b)$

c. $g'(a) = \frac{1}{f'(a)}$

d. $g'(a) = -\frac{1}{f'(b)}$

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8 Let $f(x) = x^3 + 2x - 1$. If g is the inverse of f, find $g'(2)$.

a. $\frac{1}{5}$

b. 5

c. $\frac{1}{14}$

d. 14

9 Let $f(x) = x^5 + x + 1$. What is the derivative of f^{-1} at $x = 3$?

a. $\frac{1}{6}$

b. 6

c. $\frac{1}{80}$

d. $\frac{1}{5}$

Section 4: Derivatives of Inverse Trigonometric Functions

10 What is $\frac{d}{dx}[\arcsin(3x)]$?

a. $\frac{3}{\sqrt{1-9x^2}}$

b. $\frac{1}{\sqrt{1-9x^2}}$

c. $\frac{3}{1+9x^2}$

d. $\frac{3}{\sqrt{1-3x^2}}$

11 Find $\frac{d}{dx}[\arctan(x^2)]$.

a. $\frac{2x}{1+x^4}$

b. $\frac{1}{1+x^4}$

c. $\frac{2x}{1+x^2}$

d. $\frac{2x}{\sqrt{1-x^4}}$

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12 What is the slope of the tangent line to $y = \arctan(x)$ at $x = 1$?

a. $\frac{1}{2}$

b. 1

c. $\frac{\pi}{4}$

d. $\frac{1}{\sqrt{2}}$

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